

1 Solution — GIAM 6.1

Problem 4

1.1 Problem

The “socks and shoes” rule is a mnemonic for inverting a composition. To undo “put on socks, then put on shoes” — socks first, then shoes — you must first take off your shoes, then your socks: the order of operations reverses on inversion. The same rule holds for relations. Prove that

$$(S \circ R)^{-1} = R^{-1} \circ S^{-1}.$$

1.2 Setup

Fix sets A , B , C and relations

$$R \subseteq A \times B, \quad S \subseteq B \times C.$$

For any relation $T \subseteq X \times Y$, the **inverse** is

$$T^{-1} = \{(y, x) \in Y \times X : (x, y) \in T\}.$$

For relations $R \subseteq A \times B$ and $S \subseteq B \times C$, the **composition** $S \circ R \subseteq A \times C$ is

$$S \circ R = \{(a, c) \in A \times C : \exists b \in B \text{ such that } (a, b) \in R \text{ and } (b, c) \in S\}.$$

In words, $S \circ R$ means “apply R first, then S ”.

A quick type check shows both sides of the claimed identity live in the same ambient set:

$$\begin{aligned} (S \circ R)^{-1} &\subseteq C \times A, \\ R^{-1} \circ S^{-1} &\subseteq C \times A, \end{aligned}$$

since $R^{-1} \subseteq B \times A$ and $S^{-1} \subseteq C \times B$, so $R^{-1} \circ S^{-1}$ (i.e. “ S^{-1} first, then R^{-1} ”) is well-defined.

1.3 Proof

Let $(c, a) \in C \times A$ be arbitrary. The following chain of equivalences holds.

$$\begin{aligned}
(c, a) \in (S \circ R)^{-1} &\iff (a, c) \in S \circ R && \text{(def. of inverse)} \\
&\iff \exists b \in B : (a, b) \in R \text{ and } (b, c) \in S && \text{(def. of composition)} \\
&\iff \exists b \in B : (b, a) \in R^{-1} \text{ and } (c, b) \in S^{-1} && \text{(def. of inverse, twice)} \\
&\iff \exists b \in B : (c, b) \in S^{-1} \text{ and } (b, a) \in R^{-1} && \text{(reorder the conjunction)} \\
&\iff (c, a) \in R^{-1} \circ S^{-1} && \text{(def. of composition).}
\end{aligned}$$

Each step is a logical equivalence based purely on the definitions of inverse and composition; reordering the conjunction in the fourth step is justified because logical “and” is commutative. Since the equivalence holds for every $(c, a) \in C \times A$, the two relations contain exactly the same ordered pairs. Therefore

$$(S \circ R)^{-1} = R^{-1} \circ S^{-1}. \quad \blacksquare$$

1.4 Remark — the Socks-and-Shoes Picture

Reading the proof in the original mnemonic: let R be “put on socks” (a relation from bare feet to socked feet) and S be “put on shoes” (a relation from socked feet to socked-and-shoed feet). Then $S \circ R$ is the morning routine.

The identity above says that to undo $S \circ R$, one must apply S^{-1} first (take off shoes), then R^{-1} (take off socks). Reversing the order is forced by composition itself — a *bridge* element $b \in B$ (here, “socked feet”) sits between R and S , and any inversion path through that bridge must traverse it the opposite way around. The mnemonic and the relational identity are the same statement at different levels of abstraction.